Recommendation Systems for Personalized Content Discovery

An Analysis of the Netflix Prize Dataset

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Problem Overview

The Challenge: Transitioning raw, sparse historical interaction data into actionable, personalized content discovery to reduce user decision fatigue.

The Dataset Constraints:

- **Scale:** 100+ million ratings, 480,000 users, 17,770 movies.
- **Extreme Sparsity:** 98.8% of the user-item interaction matrix is completely empty (missing data).

Core Machine Learning Objectives:

1. *Rating Prediction:* Forecasting exact 1–5 star ratings (RMSE).
2. *Relevance Ranking:* Generating highly accurate Top-10 lists (MAP@10).

Approach: Data Engineering & EDA

Exploratory Insights:

- **The Long Tail:** A small fraction of blockbusters accounts for the vast majority of ratings, while thousands of niche movies have almost none.
- **Cold-Start Problem:** High density of users with < 5 total ratings.

Engineering Solutions:

- **Memory Optimization:** Downcasted 64-bit Pandas defaults to `int32` and `int8`, reducing matrix RAM footprint to 548 MB (Parquet).
- **Smart Sampling:** Filtered the matrix to exclusively train on *Active Users* (≥ 50 ratings) to establish a dense mathematical learning signal.

Approach: Model Architectures

To evaluate different recommendation paradigms, a linear model and a non-linear deep learning model were deployed and compared.

1. Singular Value Decomposition (SVD):

- Captures linear relationships via embedding dot products.
- *Tuning:* 3-fold cross-validation Grid Search for optimal hyperparameters (`n_factors`=50, `lr`=0.005, `reg_all`=0.05).

2. Neural Collaborative Filtering (NCF):

- Captures complex, non-linear interactions via PyTorch.
- *Architecture:* 32-D continuous embedding vectors concatenated into a Multi-Layer Perceptron ($64 \rightarrow 32 \rightarrow 1$) optimized via MSE Loss.

Experimental Results

Models were evaluated strictly on the hold-out test set (`probe.txt`). A relevance threshold of ≥ 3.5 stars was used for MAP@10 calculations.

Model Architecture	**RMSE** ($\downarrow$)	**MAP@10** ($\uparrow$)
Global Mean (Baseline)	1.0628	—
User-Based CF	1.0929	0.8514
Item-Based CF	1.1585	0.8238
SVD (Tuned)	**1.0409**	**0.5573 (55.7%)**
PyTorch NCF	**1.0723**	**0.5560 (55.6%)**

Note: By utilizing the active-user filtering strategy, the Top-10 ranking precision was raised from a random baseline of $\sim$17% to over 55%.

Recommendation Examples (Qualitative Analysis)

A success/failure analysis was conducted via the custom Recommendation Generation Module.

Case Study: User 69403 (NCF Model)
The deep learning model achieved 100% precision (0 false positives) for this user's known preferences.

- *Mortal Kombat* (Pred: 4.88 — Actual: 5.0)
- *Lord of the Rings: Fellowship* (Pred: 4.54 — Actual: 4.0)
- *X-Men* (Pred: 4.21 — Actual: 4.0)
- *Lethal Weapon* (Pred: 3.99 — Actual: 4.0)

Insight: The model successfully inferred a "90s Sci-Fi/Action" latent cluster purely from the collaborative interaction matrix, without explicit metadata.

Key Insights

1. **The Criticality of Density:** Standard collaborative filtering fails on uniform random samples. Establishing a sparsity floor (≥ 50 ratings) is mandatory for extracting valid recommendations from the noise.
2. **Linear vs. Non-Linear Dominance:** For strictly predicting explicit numerical boundaries on highly sparse matrices, the linear inner-product space of SVD is exceptionally efficient and matches deep learning MLP frameworks with far less computational overhead.
3. **Rating vs. Ranking Paradox:** Optimizing a model purely for Mean Squared Error (exact star prediction) does not guarantee the best ranking order for an end-user UI.

Future Improvements

To push the boundaries of this recommendation engine in production, three architectural upgrades are proposed:

- **Bayesian Personalized Ranking (BPR):** Transitioning the neural network's loss function to pairwise learning to explicitly optimize the MAP@10 metric rather than RMSE.
- **Generalized Matrix Factorization (GMF):** Building a dual-headed network to mathematically fuse SVD dot-products with non-linear MLP layers.
- **Temporal Dynamics:** Incorporating a time-decay weight into the embedding layers to adapt as user tastes evolve over years.

Thank You.